

Figure S5

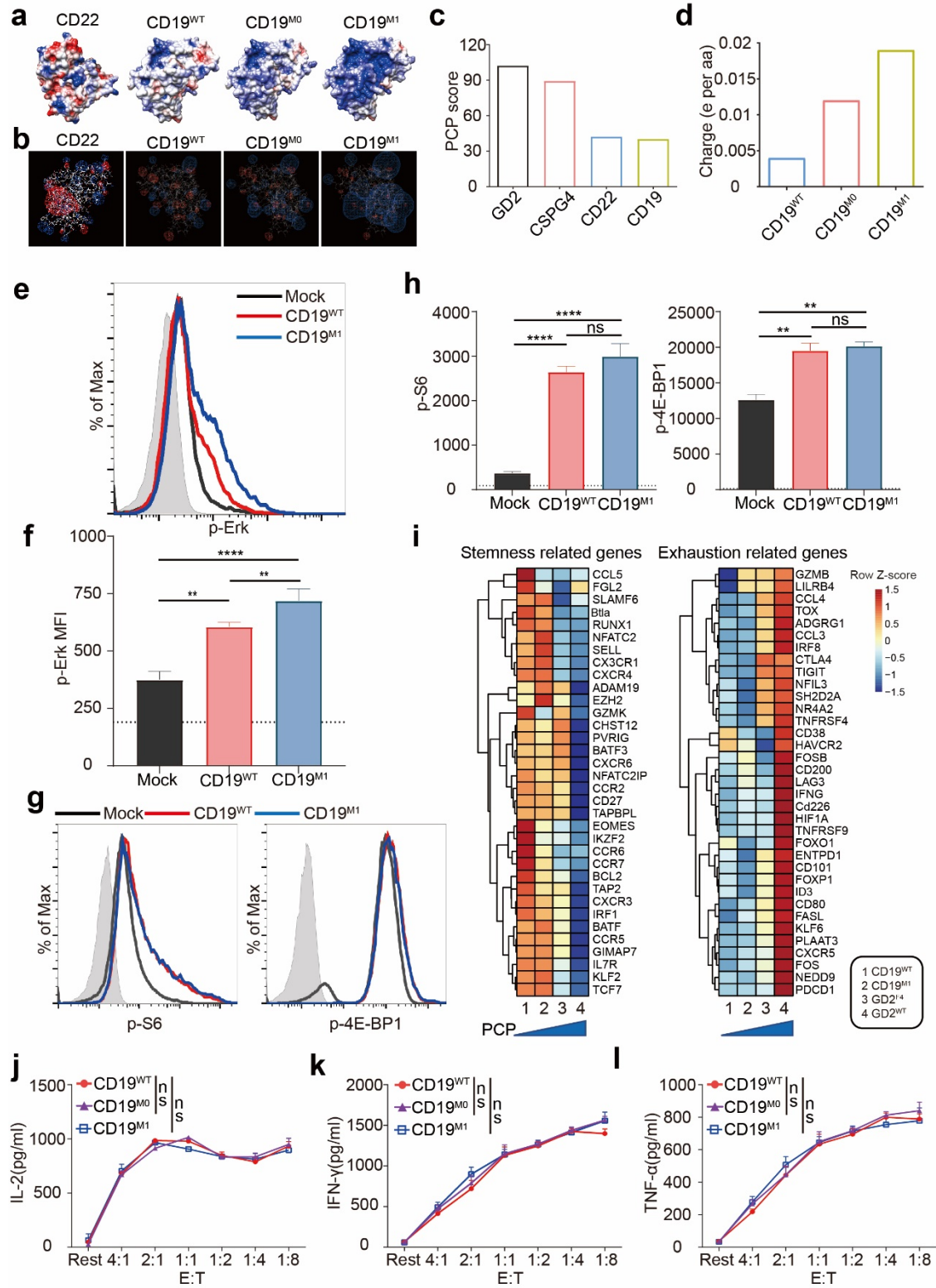


Figure S5. Tuning up PCPs on CD19 CAR surface enhances CAR-T cell fitness.

(a) Electrostatics analysis of CD22 and optimized CD19 CAR scFv constructs using APBS within UCSF Chimera. Blue, positively charged surface; red, negatively charged surface.

(b) The electrostatic potential fields of CD22 and optimized CD19 CAR scFv observed in the Swiss-PDBViewer software.

(c) The comparison of PCP scores between CD22 CAR and other CARs used in this study.

(d) Net amino acid charges of CD22 and optimized CD19 CAR scFvs at 0.15M ionic strength and pH 7.5 calculated using the Protein-Sol webserver.

(e-f) Phosphorylation of ERK in CD19^{WT} and CD19^{M1} CAR-T cells measured by FACS.

(g-h) Phosphorylation of S6 and 4E-BP1 in CD19^{WT} and CD19^{M1} CAR-T cells detected by FACS.

(i) Expressions of stemness- and exhaustion-related genes in CAR-T cells.

(j-l) Cytokine secretion assay of optimized CD19 CAR-T cells after activation. Indicated CAR-T cells were harvested and co-incubated with K562-CD19 cells at the indicated E:T ratio. The levels of IL-2 and IFN- γ were determined by ELISA.

Data are presented as means $\pm$ SEM; Comparisons were determined using unpaired student's t-tests (f, h) and two-way analysis of variance (j-l); ** $P < 0.01$; **** $P < 0.0001$; ns not significant.